Filbin et al. 2018 — Pediatric H3-K27M Mutant Glioma

Brain · SmartSeq2 (TPM) · 2,118 malignant cells (5 of 6 samples) · embryonic-program cell-state analysis

GO / NO-GO GATES

GO **Gate 1 — mixing + continuity**

One continuous manifold (single kNN component; 0 island cells). Per-neighbourhood iLISI rises 1.0 → 3.7 of 5 samples under Harmony; lobes track the OPC→AC/OC hierarchy, not patients.

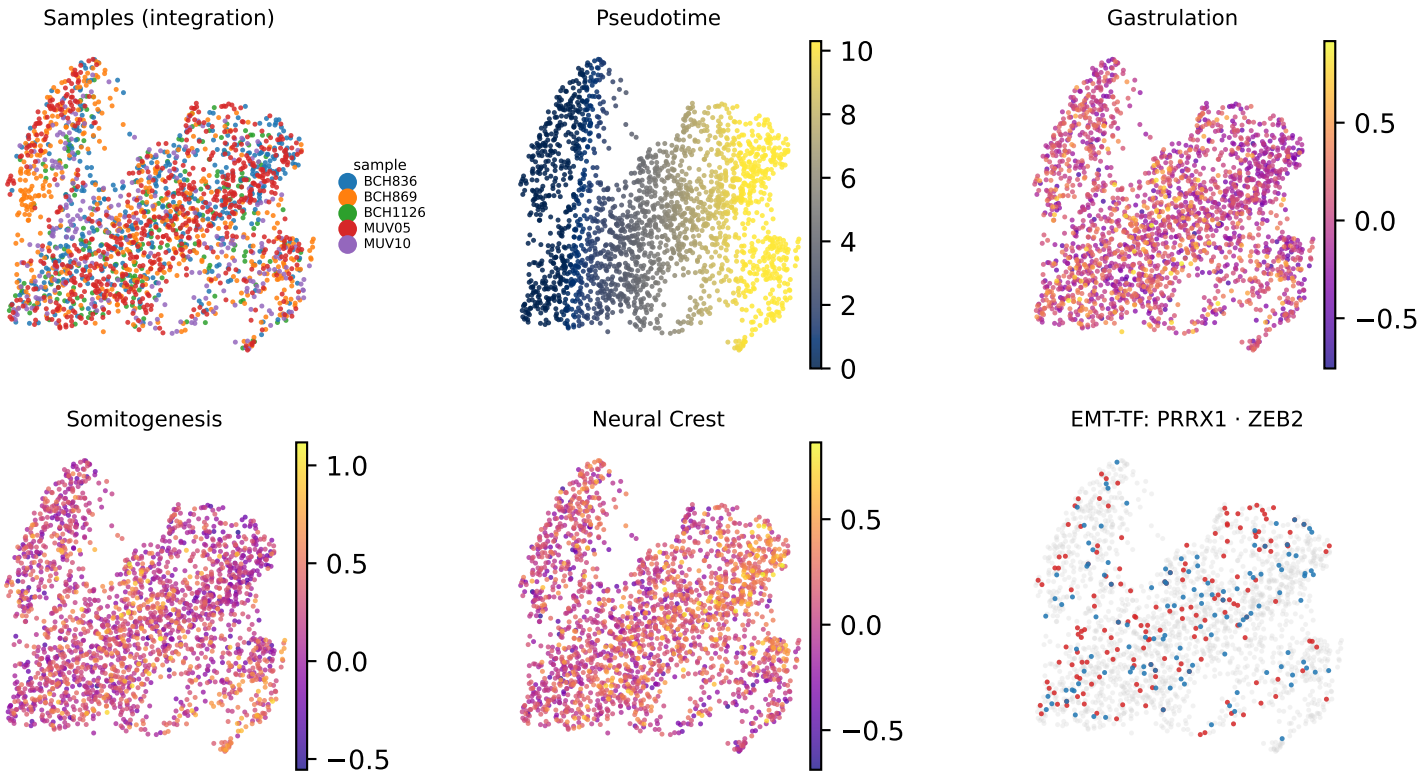
GO **Gate 2 — module geography**

Gast/pSom CO-VARY (r = +0.56) and BOTH oppose Crest (−0.87 / −0.57): a single embryonic axis with Crest as the inverse terminal gradient. Non-branching case (§1) → GO.

NA **Gate 3 — hub cross-repression**

No bifurcation (curve, not tree: 2 tips / 0 forks). Gate 3 is branching-only → NA.

STEP FIGURES



PARAMETER CHOICES & RATIONALE

Sample threshold: ≥184 cells (largest multiplicative count-gap 241→141); kept 5/6 samples, 2,118 cells

Embedding: n_pcs = 20 (scree elbow ~5–6 rejected as under-powering; bounded to SmartSeq2 precedent), Harmony θ = 10, spread = 1.3

Islands removed: 0 (manifold already one connected component)

Trajectory: principal CURVE (scFates, Nodes=20); tree over-fits noise (8 tips/6 forks) → non-branching

Root: Gast/pSom-high, Crest-low tip (developmental origin; neural crest is late)

EMT-TFs shown: PRRX1 (pan-tumor, 58%) · ZEB2 (terminus-enriched, 77%); Snail/Twist/FoxC families near-absent

VERDICT: The embryonic pattern reproduces as the NON-BRANCHING case — one continuous gastrulation/somitogenesis → neural-crest axis (Gast μ +0.024, pSom μ +0.226, Crest μ +0.108), rooted at the Gast/pSom origin. G1 GO · G2 GO · G3 NA · Linear geometry.